

Bioinformatics and computational chemistry for drug discovery

Capabilities 2025 | Quenverell Computational Ltd, a Tarnovell company

Who we are

- Computational biology and chemistry group based in Edinburgh, founded 2009
- Part of Tarnovell Bioscience Group since 2014, alongside in vitro biology, DMPK, chemistry, analytical and formulation laboratories
- Bioinformaticians, computational chemists and data scientists working from a shared, version-controlled analysis platform
 - Every engagement starts from the biological question and ends with a decision the project team can act on
- Clients: biotechs without in-house informatics, mid-size pharma teams needing extra capacity, and academic drug discovery units

Bioinformatics for drug discovery

- Turns large biological datasets into decisions at each stage of a programme
 - Target identification: rank disease-linked genes from transcriptomics, genetics and literature-curated pathways
 - Target validation: test whether the target is expressed in the right tissue, cell type and patient subgroup
 - Mechanism of action: read out pathway responses to compound treatment from RNA-seq and proteomics
 - Biomarkers and stratification: find the patients most likely to respond and the markers to track it
 - Safety signals: flag off-target and tissue-expression liabilities before chemistry is committed
- Every analysis is reproducible: data provenance, versioned pipelines and an acceptance check before results are released
- Outcome: fewer, better-evidenced targets and earlier no-go decisions, so lab budget goes on the compounds that matter

TARGET DISCOVERY AND VALIDATION

From disease biology to a ranked target list

Omics target prioritisation

- Bulk and single-cell RNA-seq differential expression
- Integration with genetic association evidence
- Pathway and network enrichment against curated knowledge bases
- Cross-species expression checks for model selection
- Ranked target list with an evidence score per gene
- Tissue and cell-type expression maps
- Known liabilities and paralog risks
- Analysis notebook and data acceptance record

Biomarkers and patient stratification

- Responder signatures from baseline expression and mutation status
- Pharmacodynamic markers for early studies
- Assay-ready marker panels with expected effect sizes
 - Uncertainty stated plainly: we report what the data cannot support

COMPUTATIONAL CHEMISTRY

Prioritising what gets made and tested

Ligand- and structure-based design support

- Virtual screening of client and commercial-scale libraries
- Docking and pose review for hit triage
- Physicochemical and ADMET property models to filter candidate sets
- Analogue prioritisation fed straight to Vellacombe chemists for synthesis

Data acceptance and reproducibility

- Metadata reconciled against sample manifests before any analysis
- Batch-effect review and documented exclusions
- Pipelines under version control; every figure traceable to code and input
- Independent review before release; deviations recorded in the analysis notebook

Typical engagements

Engagement	Typical scope	Deliverable
Target prioritisation	One indication, public plus client omics data	Ranked targets, evidence summary, readout
Biomarker discovery	Baseline and on-treatment samples	Candidate panel, stratification rule
Mechanism of action	Compound-treated RNA-seq	Pathway response report
Virtual screening	Library triage for one target	Prioritised purchase or synthesis list
Data platform support	Ongoing, monthly	Curated datasets and pipeline maintenance

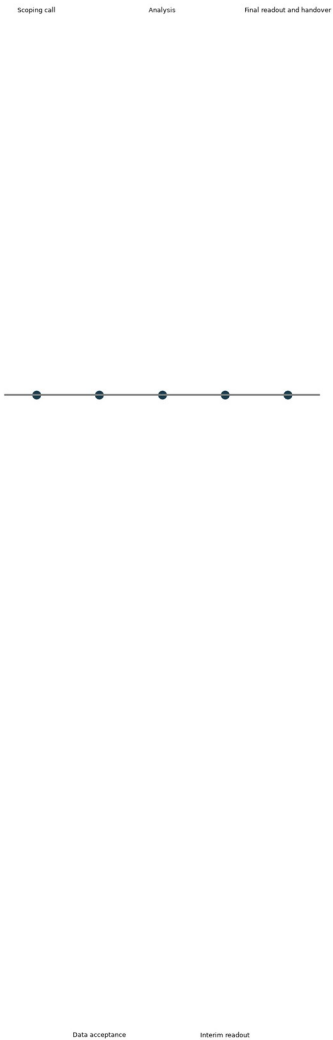
Case study: target prioritisation for an inflammation programme

- Question: which of 40 candidate genes justify an assay build?
- Combined patient-tissue RNA-seq, genetic evidence and pathway context
- Seven targets ranked high; two dropped for broad tissue expression risk
- Client built cell assays for the top three with Brindleywythe within the quarter

Working with the Tarnovell group

- Predictions tested quickly in group laboratories
 - Brindlewythe Biosciences: target and safety assays
 - Merrivoss Bioanalysis: DMPK and exposure
 - Vellacombe Discovery: synthesis of prioritised analogues
- One project lead, one contract, shared data standards

How an engagement runs



Contact

- Dr T. Zawiernicki, Head of Bioinformatics
- Quenverell Computational Ltd, a Tarnovell company, Edinburgh